Set

In [2]:
```
s = {}
s
```

Out[2]:
```
{}
```

In [3]:
```
type(s)
```

Out[3]:
```
dict
```

In [4]:
```
s1 = set()
s1
```

Out[4]:
```
set()
```

In [5]:
```
type(s1)
```

Out[5]:
```
set
```

In [6]:
```
s2 = {100,25,50,5,20,30}
s2
```

Out[6]:
```
{5, 20, 25, 30, 50, 100}
```

In [7]:
```
len(s2)
```

Out[7]:
```
6
```

In [8]:
```
s2.add(15)
```

In [9]:
```
s2
```

Out[9]:
```
{5, 15, 20, 25, 30, 50, 100}
```

In [10]:
```
s3 = s2.copy()
```

In [11]:
```
s3
```

Out[11]:
```
{5, 15, 20, 25, 30, 50, 100}
```

In [12]:
```
print(s1)
print(s2)
print(s3)
```

```
set()
{50, 100, 5, 20, 25, 30, 15}
{50, 100, 5, 20, 25, 30, 15}
```

In [13]:
```
s2
```

Out[13]:
```
{5, 15, 20, 25, 30, 50, 100}
```

In [14]:
```
s2.pop()
```

Out[14]:
```
50
```

In [15]:
```
s2.discard(100)
s2
```

Out[15]:
```
{5, 15, 20, 25, 30}
```

In [16]:
```
s2
```

Out[16]:
```
{5, 15, 20, 25, 30}
```

In [17]:
```
s2.add(25)
s2
```

Out[17]:
```
{5, 15, 20, 25, 30}
```

Set operations

UNION

In [20]:
```
a1 = {1,2,3,4,5}
b1 = {4,5,6,7,8}
c1 = {8,9,10}
```

In [21]:
```
a1.union(b1)
```

Out[21]:
```
{1, 2, 3, 4, 5, 6, 7, 8}
```

In [22]:
```
a1.union(b1,c1)
```

Out[22]:
```
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

In [23]:
```
a1 | b1
```

Out[23]:
```
{1, 2, 3, 4, 5, 6, 7, 8}
```

In [24]:
```
b1 | c1
```

Out[24]:
```
{4, 5, 6, 7, 8, 9, 10}
```

In [25]:
```
a1|b1|c1
```

Out[25]:
```
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

intersection

In [27]:
```
print(a1)
print(b1)
print(c1)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

In [28]:
```
a1.intersection(b1)
```

Out[28]:
```
{4, 5}
```

In [29]:
```
a1.intersection(c1)
```

Out[29]:
```
set()
```

In [30]:
```
a1 & b1 & c1
```

Out[30]:
```
set()
```

In [31]:
```
a1 & b1
```

Out[31]:
```
{4, 5}
```

In [32]:
```
b1 & c1
```

Out[32]:
```
{8}
```

In [33]:
```
a1 & c1
```

Out[33]:
```
set()
```

In [34]:
```
a1 = {1,2,3,4,5}
b1 = {4,5,6,7,8}
c1 = {8,9,10}
```

In [35]:
```
a1.difference(b1)
```

Out[35]:
```
{1, 2, 3}
```

In [36]:
```
a1.symmetric_difference(b1)
```

Out[36]:
```
{1, 2, 3, 6, 7, 8}
```

In [37]:
```
a1 ^ b1
```

Out[37]:
```
{1, 2, 3, 6, 7, 8}
```

In [38]:
```
a1.symmetric_difference_update(b1)
```

In [39]:
```
a1
```

Out[39]:
```
{1, 2, 3, 6, 7, 8}
```

In [40]:
```
a1.difference(b1)
```

Out[40]:
```
{1, 2, 3}
```

In [41]:
```
a1
```

Out[41]:
```
{1, 2, 3, 6, 7, 8}
```

In [42]:
```
a1.difference_update(b1)
a1
```

Out[42]:
```
{1, 2, 3}
```

In [43]:
```
print(a1)
print(b1)
print(c1)
```

```
{1, 2, 3}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

superset / subset / disjoint

In [45]:
```
s5 = {1,2,3,4,5,6,7,8,9}
s6 = {3,4,5,6,7,8}
s7 = {10,20,30,40}
```

In [46]:
```
s5.issuperset(s6)
```

Out[46]:
```
True
```

In [47]:
```
s6.issubset(s5)
```

Out[47]:
```
True
```

In [48]:
```
s6.issuperset(s5)
```

Out[48]:
```
False
```

In [49]:
```
s7.isdisjoint(s5)
```

Out[49]:
```
True
```

In [50]:
```
s7.isdisjoint(s6)
```

Out[50]:
```
True
```

In [51]:
```
s5.isdisjoint(s6)
```

```
Out[51]: False
```